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Spatially Aware k-Nearest-Neighbours Regression Methods

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Abstract

The inherent presence of spatial autocorrelation in spatial data necessitates special consideration when working with geographic data. Many algorithms and machine learning techniques used for spatial data prediction, including k-Nearest Neighbours, do not account for spatial autocorrelation. This study proposes Spatially Aware k-Nearest Neighbours (SA-kNN), which predicts values at an unsampled location by positively weighting neighbours that contribute more unique information than others. We also propose adding regression-kriging to such algorithms to capture residual spatial structure. SA-kNN and some SA-kNN variants are then applied to house sale data from Franklin County, Ohio, and compared with several other spatial learning approaches.

1 Introduction

A fundamental problem in geography is estimating a value at an unsampled location, given a surrounding sample. Solving this problem provides utility in the analysis of broad classes of spatial data, including (but not limited to) housing markets (Chen, Lee, and Chun 2026; Baldominos et al. 2018), demographic information (Openshaw and Taylor 1979; Griffith et al. 2022), and, increasingly, climate or environmental data (Hengl et al. 2018; Foresti et al. 2010). Many techniques have been developed to attempt to solve this problem, but each has its own unique advantages and drawbacks (Cressie 1993; Bivand, Pebesma, and Gómez-Rubio 2013). Machine learning or other nonparametric methods such as k-Nearest Neighbours (kNN) are popular for this problem, owing to their simplicity and flexibility (Hastie et al. 2021; Peterson 2009).

However, it is well-known that spatial data has a distinctive property: spatial autocorrelation. Per Tobler's First Law, "Everything is related to everything else, but near things are more related than distant things" (Tobler 1970). As a consequence, spatial observations are not independent, which affects the results obtained when one uses machine learning methods that assume independence on them. In particular, standard kNN treats all neighbours as if they contribute equally unique information, but when neighbours are spatially clustered and correlated, this is not necessarily true. Work on effective sample size shows that correlated observations reduce the true information content of a sample (Griffith 2005; Acosta, Vallejos, and Griffith 2018; Vallejos and Osorio 2014). As an example, imagine trying to estimate the price of a house in a neighbourhood where all sampled observations are new houses on the same block, all built by the same developer. These houses will likely have similar prices, and although they are a helpful indicator for the overall estimation, they contribute redundant information, which kNN can fail to detect; if the k neighbours include all of these houses, they may 'pollute' the pool of data for the prediction.

Several solutions to this sub-problem have been proposed; some notable methods weight neighbours by distance or add locational information (Dudani 1976; Atkinson 2004; Bal-

dominos et al. 2018). In this paper, we outline Spatially Aware k-Nearest Neighbours (SA-kNN), which modifies the weighting step of the algorithm by accounting for redundancy among neighbours. Specifically, neighbours that are more correlated with other selected neighbours receive lower weights. SA-kNN is then combined with the recently proposed SE-KNN blended-distance selection (Chen, Lee, and Chun 2026), which ensures that selected neighbours are geometrically distributed around the query rather than spatially clustered. Finally, a regression-kriging correction stage is added, which interpolates the spatial residual field left by the feature-based prediction and adds it back to the final estimate. We apply these methods to residential sales data from Franklin County, Ohio, and compare performance against standard methods across multiple metrics.

The rest of this paper is organized as follows. Section 2 reviews the literature on spatial autocorrelation, spatial interpolation, and spatially aware prediction methods. Section 3 presents the SA-kNN algorithm, its combination with SE-KNN, the regression-kriging correction, and a description of the study area and data. Section 4 presents results and discusses key findings; we then conclude.

2 Literature Review

2.1 Spatial autocorrelation in GIScience

Spatial autocorrelation refers to the degree to which a variable's values at nearby locations resemble each other more than values at distant locations. This property is pervasive in spatial data and follows directly from Tobler's First Law of Geography: "everything is related to everything else, but near things are more related than distant things" (Tobler 1970). There is no universal measure of spatial autocorrelation.

Historical development. The formal study of spatial autocorrelation emerged from the need to evaluate the distribution of spatial processes, such as disease. One such early example is John Snow's mapping of the 1854 Soho cholera outbreak in London (Snow 1855): by plotting deaths as points on a street map and tracing their clustering to the Broad Street water pump, Snow showed that spatial proximity alone can serve as evidence for a hypothesis.

Patrick Moran (1950) introduced the first rigorous mathematical test for spatial autocorrelation, Moran's I , which we will soon develop in more detail. Geary (1954) proposed an alternative measure, Geary's c , based on squared differences between contiguous pairs, while Clark and Evans (1954) introduced the nearest-neighbour index for point patterns. Cliff and Ord (1973) then derived a full distributional theory for global spatial autocorrelation statistics. Global measures of spatial autocorrelation summarize the entire study region

with a single value, thereby assuming a spatially stationary process in which the same autocorrelation structure holds everywhere.

The 1990s brought a shift toward *local* spatial statistics. Getis and Ord (1992) proposed the G_i^* statistic to identify local hot spots and cold spots, i.e., locations where high or low values cluster significantly beyond what would be expected. Anselin (1995) introduced Local Indicators of Spatial Association (LISA), decomposing the global Moran's I into location-specific contributions and enabling classification of each observation as a spatial cluster, spatial outlier, or neither. This transition from global to local models aligns with shifts in GIScience that center the local context for geographic processes, adapting to spatial heterogeneity (Goodchild 2004). This shift was the precursor to spatially varying coefficient models such as Geographically Weighted Regression and, ultimately, to the geographically explicit machine learning methods reviewed in Section 2.4.

In mathematical terms, a random field $\{Z(\mathbf{s}) : \mathbf{s} \in D \subset \mathbb{R}^2\}$ exhibits positive spatial autocorrelation if $\text{Cov}(Z(\mathbf{s}_i), Z(\mathbf{s}_j)) > 0$ for nearby pairs $\mathbf{s}_i, \mathbf{s}_j$ (with covariance generally decreasing as $\|\mathbf{s}_i - \mathbf{s}_j\|$ increases) (Cressie 1993; Cliff and Ord 1981).

The most widely used summary statistic for spatial autocorrelation is Moran's I (Moran 1950; Cliff and Ord 1973):

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij} (z_i - \bar{z})(z_j - \bar{z})}{\sum_i (z_i - \bar{z})^2}, \quad (1)$$

where $z_i = Z(\mathbf{s}_i)$, $\bar{z}$ is the sample mean, and w_{ij} are spatial weights. Popular spatial weights include binary contiguity (via a simple adjacency matrix) or inverse-distance. $I \approx 0$ indicates no significant spatial autocorrelation; $I \gg 0$ indicates clustering of similar values. Furthermore, in the context of prediction, Moran's I computed on the residuals of a model measures the degree to which prediction errors are spatially clustered. In particular, a persistent positive residual Moran's I indicates that the model has failed to fully capture the spatial pattern.

One practically important consequence of spatial autocorrelation is that it reduces the effective sample size of a spatial dataset. Griffith (2005) shows that for a sample of size n with average pairwise correlation r , the effective sample size is approximately

$$n_{\text{eff}} \approx \frac{n}{1 + (n-1)r}$$

. This is effectively a formal version of the illustrative example in the introduction about estimating from a sample of houses all built by the same developer, which motivates the redundancy-weighting for SA-kNN.

Nearness and scale effects. Tobler's First Law asserts that "near things are more related," but what it means to be near is itself a methodological choice that encodes assumptions

about how a phenomenon is structured. Part of this is addressed by models that use non-geographic information - such as those like SE-SA-kNN (Section 3.3) that look for neighbours in feature space instead of geographic space. However, we must still include some encoding of physical geography if we want to address spatial autocorrelation.

Furthermore, Openshaw (1984) demonstrated that spatial autocorrelation statistics are sensitive to the scale and configuration of the spatial units of analysis. Aggregating point data into zones with different boundaries can produce dramatically different patterns that are merely an artifact of the aggregation rather than a property of the underlying process. In this paper, we do not apply our models to aggregated data (and so escape this scale problem), but analogous scale-dependencies appear in our selection of models. In particular, the spatial bandwidth h in the regression-kriging correction (Section 3.4) and the $k = 8$ neighbourhood used for computing residual Moran's I are choices that shape the degree of residual spatial structure. We cannot claim to have a neutral methodology (Harley 1989).

2.2 Spatial interpolation: deterministic and stochastic approaches

The problem of predicting values at unsampled locations from a surrounding sample has produced two fundamentally different families of methods.

Deterministic methods. The earliest formalized interpolation method is Thiessen polygons, a precursor to nearest-neighbour regression (Thiessen 1911): each unsampled location is assigned the value of its single nearest observation, partitioning the study area into (Voronoi) polygons. Shepard (1968) introduced Inverse Distance Weighting (IDW) as a smoother alternative: the prediction is a weighted mean of surrounding observations, with weights proportional to d^{-p} for some power p . Both methods are deterministic, as given the data, the predicted value is fully determined. There is no representation of uncertainty.

Stochastic methods. Krige (1951)'s empirical observation that ore grade at South African gold mines varied with geographic distance led to a basic theory of kriging. Matheron (1963) formalized this into the theory of regionalized variables. Kriging uses the empirical (data-based) variogram (relationship between downside variance and geography) to derive optimal linear prediction weights that minimize mean squared error (MSE) and return a prediction variance and a point estimate. Kriging is therefore stochastic: it acknowledges that spatial prediction is uncertain and quantifies its confidence. In contrast, a smooth IDW surface appears authoritative, with its values presented as if they were precise measurements - but this appearance is partly an artifact of the method. The geographic visualisations in Section 4 should be read with this in mind.

2.3 k-Nearest Neighbours

k-Nearest Neighbours (kNN) was first proposed (as a classification algorithm) by Fix and Hodges (1951) in an unpublished technical report; Cover and Hart (1967) wrote up its theoretical foundations. kNN for continuous prediction (i.e., estimating a response rather than assigning a class label) follows the same principle, and has been applied to spatial prediction tasks including land cover mapping (Atkinson 2004), property valuation (Baldominos et al. 2018), and flood-prone area delineation (Razavi-Termeh et al. 2024).

The primordial limitation of kNN when applied to spatial contexts is its assumption of observation independence, as standard kNN treats each training point as independently and identically distributed. But Cliff and Ord (1973) showed that spatially autocorrelated observations reduce the effective degrees of freedom and invalidate the theoretical foundation of inference procedures that assume independence. More practically, when k nearest neighbours in feature space are also geographically proximate (as spatial autocorrelation implies they often will be) they carry redundant information. In a probabilistic sense, they have been drawn from the same local spatial process and will tend to share similar values for reasons unrelated to their individual attributes. This redundancy is the central motivation for SA-kNN.

k-Nearest Neighbours (kNN) is a nonparametric prediction algorithm that estimates a value at query point $\mathbf{q}$ as the (weighted) mean of the k most similar training observations (Hastie et al. 2021; Peterson 2009). In its standard form, similarity is measured by Euclidean distance in feature space, and the prediction is an unweighted average:

$$\hat{y}(\mathbf{q}) = \frac{1}{k} \sum_{i=1}^k y_i, \quad \mathbf{x}_{(i)} \in \mathcal{N}_k(\mathbf{q}), \quad (2)$$

where $\mathcal{N}_k(\mathbf{q})$ is the set of k nearest neighbours in feature space. A natural extension, Inverse Distance Weighting (IDW) kNN (Dudani 1976), weights each neighbour by the reciprocal of its feature-space distance:

$$\hat{y}(\mathbf{q}) = \frac{\sum_{i=1}^k d_i^{-1} y_i}{\sum_{i=1}^k d_i^{-1}}, \quad d_i = \|\mathbf{x}_{(i)} - \mathbf{q}\|. \quad (3)$$

IDW assigns higher weight to closer neighbours, but still does not differentiate between two close neighbours that share redundant information and two close neighbours that provide complementary information.

2.4 Related spatially aware methods

Several approaches have been proposed to make prediction algorithms spatially “aware” (in a general sense, not in an artificial general intelligence sense). We consider some of them

here.

Geographically Weighted Regression (GWR). GWR (Fotheringham, Brunsdon, and Charlton 2002) fits a separate linear model at each prediction location, using a kernel function to downweight distant observations. This captures spatial non-stationarity (i.e. coefficients varying over space) but relies on linear relationships and can be computationally expensive for large datasets.

Spatially explicit machine learning (ML). A growing body of work incorporates spatial features (coordinates, spatial lags, eigenvectors) into standard ML models to induce spatial awareness (Liu, Kounadi, and Zurita-Milla 2022; Nikparvar and Thill 2021; Kopczewska 2022). In general, these methods modify the features but not the prediction or weighting step.

SE-KNN (Chen et al., 2026). The most directly related method is SE-KNN (Chen, Lee, and Chun 2026), which selects neighbours using a blended distance metric:

$$d'_{ij} = (1 - \lambda) \|\mathbf{x}_i - \mathbf{x}_j\| + \lambda \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|, \quad \lambda \in [0, 1], \quad (4)$$

where $\boldsymbol{\mu}_i$ is the mean feature vector of a spatial neighbourhood of $\mathbf{x}_i$. By blending feature-space and spatial-neighbourhood distances, SE-KNN ensures that neighbours are spatially distributed around the query point, reducing systematic under- or over-prediction at the tails of the spatial distribution. It is worth noting that SE-KNN makes no modification to the weighting step, treating all selected neighbours equally. SA-kNN can then be viewed as a complementary approach in this sense, where we modify the weights rather than the selection criterion.

Regression-kriging and spatial error models. In geostatistics, regression-kriging decomposes prediction into a regression trend and a kriging interpolation of the residual spatial field (Odeh, McBratney, and Chittleborough 1995; Hengl, Heuvelink, and Rossiter 2007). The idea is that residuals from regression carry spatially structured information (i.e. their Moran's I is non-zero) that can be modelled separately and subsequently added back to improve predictions. This motivates the regression-kriging correction stage introduced in Section 3.4.

Spatial GP regression. Gaussian Process regression with a spatial kernel (kriging) is a classical geostatistical approach to spatial prediction (Cressie 1993). Recent variants such as geographic GP (Jiao and Tao 2025) integrate feature-space similarity, as in SE-KNN. However, these methods require explicit parametric covariance assumptions and scale poorly to large datasets.

GeoAI and interpretability. The rise of GeoAI has included the application of deep learning, neural networks, and transformer architectures to spatial prediction (Janowicz et al. 2020; Li 2020). These models can often capture highly non-linear spatial representations and outperform classical spatial statistics. However, there is a tradeoff in interpretability. Roscher et al. (2020) argue that a model whose actions cannot be understood cannot be validated against prior domain knowledge, corrected when it makes systematically wrong predictions in specific geographic contexts, or held accountable in policy applications. The inscrutability of some prediction approaches means they can act as ‘black box’ models, especially if they are proprietary methods. The methods developed in this paper are, as a response to this idea, interpretable and reproducible by design.

3 Methodology

3.1 Problem setup

Let $\{(\mathbf{s}_i, \mathbf{x}_i, y_i)\}_{i=1}^n$ denote a training sample, where $\mathbf{s}_i \in \mathbb{R}^2$ is the spatial location of observation i , $\mathbf{x}_i \in \mathbb{R}^p$ is a p -dimensional feature vector, and $y_i \in \mathbb{R}$ is the response. Given a query point $(\mathbf{s}_q, \mathbf{x}_q)$, the goal is to predict y_q .

3.2 SA-kNN algorithm

SA-kNN selects the k nearest neighbours of $\mathbf{x}_q$ in feature space (as in standard kNN) but modifies the prediction weights to account for spatial redundancy among the selected neighbours. If two selected neighbours i and j are spatially close to each other, they collectively carry less unique information than two spatially distant neighbours. SA-kNN does this by computing a redundancy score for each neighbour.

Let $C_{ij} \in [0, 1]$ be a measure of spatial correlation between neighbours i and j . The redundancy score of neighbour i is defined as the mean correlation it shares with all other neighbours in the set:

$$r_i = \frac{1}{k-1} \sum_{j \neq i} C_{ij}. \quad (5)$$

A high r_i indicates that neighbour i is correlated with many other selected neighbours and thus contributes less unique information. The effective weight of neighbour i is then

$$w_i^{\text{eff}} = \frac{d_i^{-1}(1 - r_i)}{\sum_{j=1}^k d_j^{-1}(1 - r_j)}, \quad (6)$$

where $d_i = \|\mathbf{x}_i - \mathbf{x}_q\|$ is the feature-space distance. The prediction is the weighted average

$$\hat{y}_q^{(1)} = \sum_{i=1}^k w_i^{\text{eff}} y_i. \quad (7)$$

Neighbours with high feature-space distance *or* high spatial correlation with other neighbours are thus downweighted.

Choice of C_{ij} . In this paper, we use a Gaussian proxy and approximate spatial correlation by a function of geographic distance:

$$C_{ij} = \exp\left(-\frac{\|\mathbf{s}_i - \mathbf{s}_j\|^2}{2h^2}\right), \quad (8)$$

where h is a bandwidth parameter, which we set to the median inter-neighbour geographic distance. Given the bandwidth choice being fixed, this approximation is otherwise parameter-free. As we see in Section 4.4, this formulation causes SA-kNN to become similar to IDW.

3.3 Combination with SE-KNN: SE-SA-kNN

SE-KNN (Chen, Lee, and Chun 2026) and SA-kNN address different aspects of the kNN algorithm: SE-KNN modifies neighbour selection, while SA-kNN modifies neighbour weighting. It is not unnatural to try a combination, which we call SE-SA-kNN. This algorithm uses SE-KNN's blended distance to select the k neighbours, then applies SA-kNN's redundancy weights for prediction. Our motivation is thus: SE-KNN's blended selection creates more diverse neighbourhood sets than just feature-space selection. This geographic diversity gives the redundancy correction greater leverage, since there is more variance in the inter-neighbour spatial correlations for the weighting step to exploit. In contrast, when neighbours are selected by feature space alone, they tend to cluster geographically (due to spatial autocorrelation), causing the $(1 - r_i)$ correction to apply roughly uniformly across the neighbour set.

3.4 Regression-kriging correction

Even after feature-based SE-SA-kNN prediction, residuals can often exhibit positive spatial autocorrelation. This is a result of spatially structured effects (in the case of housing data: neighbourhood quality, proximity to amenities, market conditions) not fully captured by the predictors. A regression-kriging stage explicitly models this residual spatial field and modifies the SE-SA-kNN prediction.

We use a standard approach (Odeh, McBratney, and Chittleborough 1995; Hengl, Heuvelink, and Rossiter 2007); we decompose the prediction target into a “trend” component (the feature-

based SE-SA-kNN estimate) and a spatially correlated residual. Then, for each training point i , compute a 5-fold cross-validated prediction $\tilde{y}_i$ using the SE-SA-kNN model, yielding an approximately unbiased residual

$$e_i = y_i - \tilde{y}_i. \quad (9)$$

For a query point $(\mathbf{s}_q, \mathbf{x}_q)$, the spatial correction is a Gaussian-kernel-weighted average of the training residuals:

$$\delta(\mathbf{s}_q) = \frac{\sum_{i=1}^n \exp\left(-\frac{\|\mathbf{s}_q - \mathbf{s}_i\|^2}{2h^2}\right) e_i}{\sum_{i=1}^n \exp\left(-\frac{\|\mathbf{s}_q - \mathbf{s}_i\|^2}{2h^2}\right)}, \quad (10)$$

where the bandwidth h is set to a parameter h_{frac} times the mean distance to the 10th geographic nearest neighbour in the training set, with h_{frac} being tuned on the validation set. The final prediction is then

$$\hat{y}_q = \hat{y}_q^{(1)} + \delta(\mathbf{s}_q), \quad (11)$$

where $\hat{y}_q^{(1)}$ is the SE-SA-kNN prediction (7).

A detail worth noting is the use of cross-validated residuals rather than in-sample residuals. In-sample kNN residuals are biased toward zero because each training point appears in its own neighbourhood. Cross-validated kNN residuals are approximately unbiased and therefore ‘nice to play with’.

The RegKrig-SE-SA-kNN algorithm. For a query $(\mathbf{s}_q, \mathbf{x}_q)$:

1. Find the blended distance d'_{ij} (4) from $\mathbf{x}_q$ to all training points; select the k nearest.
2. Compute the Gaussian proximity proxy C_{ij} (8) for all pairs of selected neighbours.
3. Compute redundancy scores r_i (5) and effective weights w_i^{eff} (6).
4. Compute $\hat{y}_q^{(1)} = \sum_i w_i^{\text{eff}} y_i$.
5. Add the regression-kriging correction $\delta(\mathbf{s}_q)$ (10).
6. Return $\hat{y}_q = \hat{y}_q^{(1)} + \delta(\mathbf{s}_q)$.

4 Results and Discussion

4.1 Study area and data

The dataset consists of residential property sales records from Franklin County, Ohio (the Columbus metropolitan area), obtained from the Franklin County Auditor’s Office. We re-

strict to calendar year 2019 to align with Chen, Lee, and Chun (2026). After filtering to arms-length sales only by excluding validity codes indicating non-market transactions (related-party, distressed, abnormal financing, non-usable deed type, unfinished new builds, and multi-parcel sales) and removing records with missing dwelling attributes, missing geographic coordinates, unrealistic dwelling sizes (less than 200 or more than 10,000 square feet), lots over five acres, and zero or missing bedroom counts, we retain $N = 12,376$ observations. Sales prices in the bottom 5th percentile are further excluded. The resulting sample is split randomly into training (64%), validation (16%), and test (20%) sets. These choices mimic the stated methodology of Chen, Lee, and Chun (2026).

Geographic coordinates are taken from the parcel centroid CSV provided by the Franklin County Auditor, in the Ohio State Plane South projection (WKID 3735, US Survey Feet). Feature-space distance is computed separately in standardized feature units. Figure 1 shows the spatial distribution of the 12,376 sale prices across Franklin County. A pronounced spatial stratification is evident: lower-value properties (\$33,000–\$100,000) are concentrated in inner-city Columbus neighbourhoods, while mid-range properties (\$100,000–\$200,000) form a broad suburban ring. Higher-value properties are predominantly located in the northern and eastern suburbs, with properties exceeding \$750,000 almost exclusively in the north. This price gradient provides substantial spatial structure for the prediction models to capture.

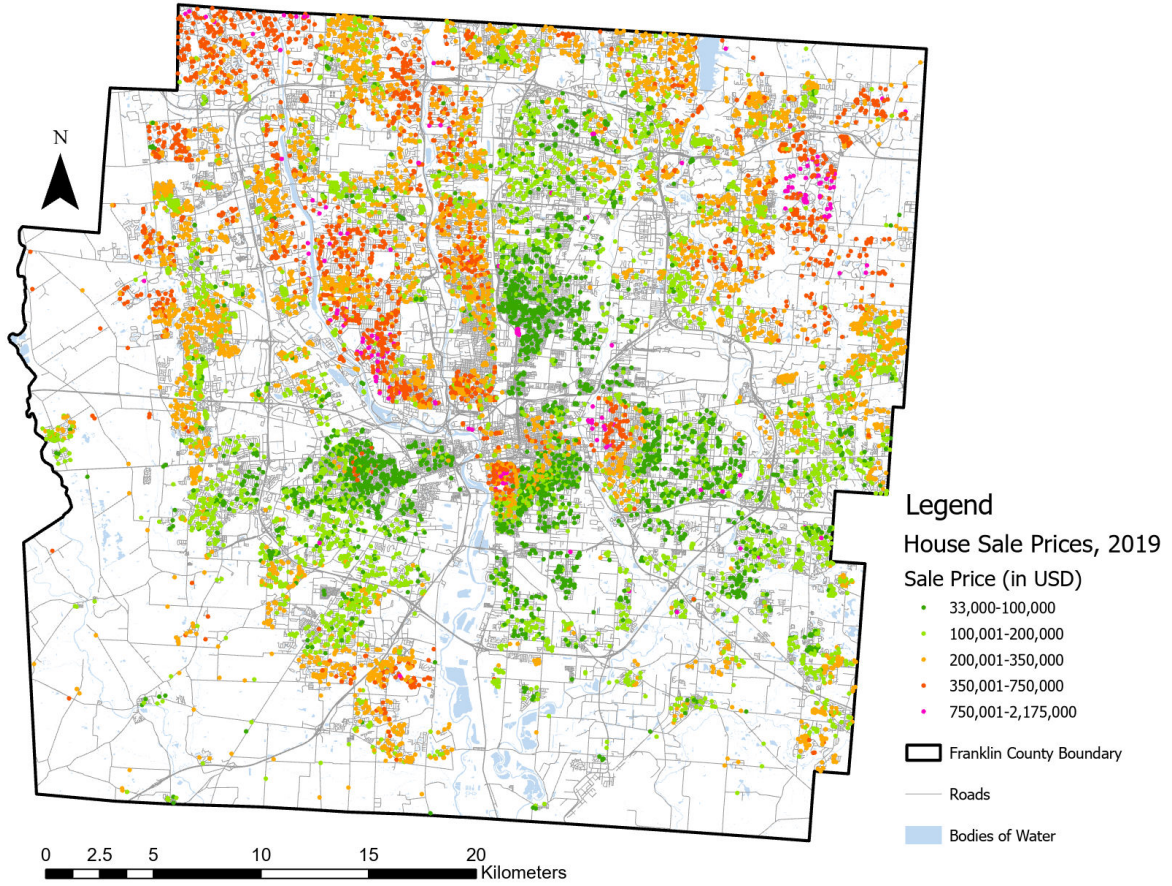


Figure 1: Spatial distribution of 12,376 arms-length residential sale prices in Franklin County, Ohio (2019). Points are coloured by five price classes.

Features. We again follow the methodology of Chen, Lee, and Chun (2026). After computing Pearson r with $\log(\text{PRICE})$ on the training set, 11 dwelling attributes with $|r| \geq 0.1$ are retained: total rooms, bedrooms (RMBED), full bathrooms (FIXBATH), half bathrooms (FIX-HALF), number of stories (STORIES_NUM), total fixtures (FIXTOT), additional fixtures (FIX-ADDL), above-grade finished living area in square feet (SFLA), condition grade (COND_ENC and GRADE_ENC), and house age (HOUSE_AGE). All features are standardized to zero mean and unit variance before any distance computations. The response variable is $\log(\text{PRICE})$; predictions are back-transformed using $\exp(\hat{y})$.

Why do we use log-price? Chen et al. do not claim to use log-price values, but we diverged from their methodology here as we found their results irreproducible. House prices are right-skewed: a small fraction of high-value properties inflate the raw-price mean and increase the leverage of outliers for distance-weighted algorithms. Log-transforming the price compresses this skew and makes the feature-to-price relationship closer to linear. All experiments use log-price, and all reported metrics are back-transformed to USD.

Discrepancies with Chen, Lee, and Chun (2026). Chen et al. also use Franklin County 2019 residential sales, but their sample contains approximately 14,000 observations, about 1,600 more than ours. We believe this discrepancy arises from differences in validity-code filtering, but cannot be sure; the details of the exact source of their data are unclear, as well as the exact data processing steps undertaken. We therefore also have to implement SE-KNN locally from their published algorithm description, as no public codebase is available.

4.2 Implementation and hyperparameters

All six models are implemented in Python using `scikit-learn` for nearest neighbour queries and `numpy / scipy` for matrix operations. Parameters (k and, for SE variants, λ ; for RegKrig, additionally the bandwidth fraction h_{frac}) are tuned on the validation set by grid search: $k \in \{2, \dots, 14\}$, $\lambda \in \{0.5, 0.6, \dots, 0.9\}$, and $h_{\text{frac}} \in \{0.5, 1.0, 2.0, 5.0\}$. Optimal parameters are then fixed, and all metrics are reported on the test set. We report root mean squared error (RMSE), mean absolute error (MAE), R^2 , and Moran’s I of residuals, the latter of which is computed using spatial kNN weights with $k = 8$.

4.3 Model performance

Test-set metrics for all six models are reported in Table 1. The progression from standard kNN through the full RegKrig-SE-SA-kNN illustrates the contribution of each modelling component. Actual-versus-predicted scatter plots for all six models appear in Figure 2. Figures 3 and 4 provide geographic views of the RegKrig-SE-SA-kNN predictions and absolute prediction errors across the county.

Table 1: Test-set prediction metrics (log-price, back-transformed to USD). Best values in each metric are **bold**. $N_{\text{test}} = 2,476$.

Method	RMSE	MAE	R^2	Moran’s I
Standard kNN ($k = 2$)	\$381,726	\$76,629	0.6420	0.1347
IDW kNN ($k = 7$)	\$220,621	\$62,903	0.8804	0.1627
SA-kNN ($k = 6$)	\$219,535	\$63,897	0.8816	0.1646
SE-KNN ($\lambda = 0.6$, $k = 3$)	\$276,184	\$57,476	0.8126	0.1351
SE-SA-kNN ($\lambda = 0.6$, $k = 6$)	\$211,404	\$52,606	0.8902	0.1528
RegKrig-SE-SA-kNN ($\lambda = 0.8$, $k = 12$, $h = 1.0$)	\$208,930	\$48,201	0.8928	0.1357

The most striking pattern in Table 1 is the large performance gap between feature-space-only methods (standard kNN, IDW, SA-kNN) and the SE-selection variants. Standard kNN achieves a MAE of \$76,629; IDW and SA-kNN improve modestly to \$62,903 and \$63,897 respectively, and remain approximately equivalent to each other (see Section 4.4). SE-KNN (\$57,476) marks a substantial increase, replicating (on a different scale) the finding of Chen,

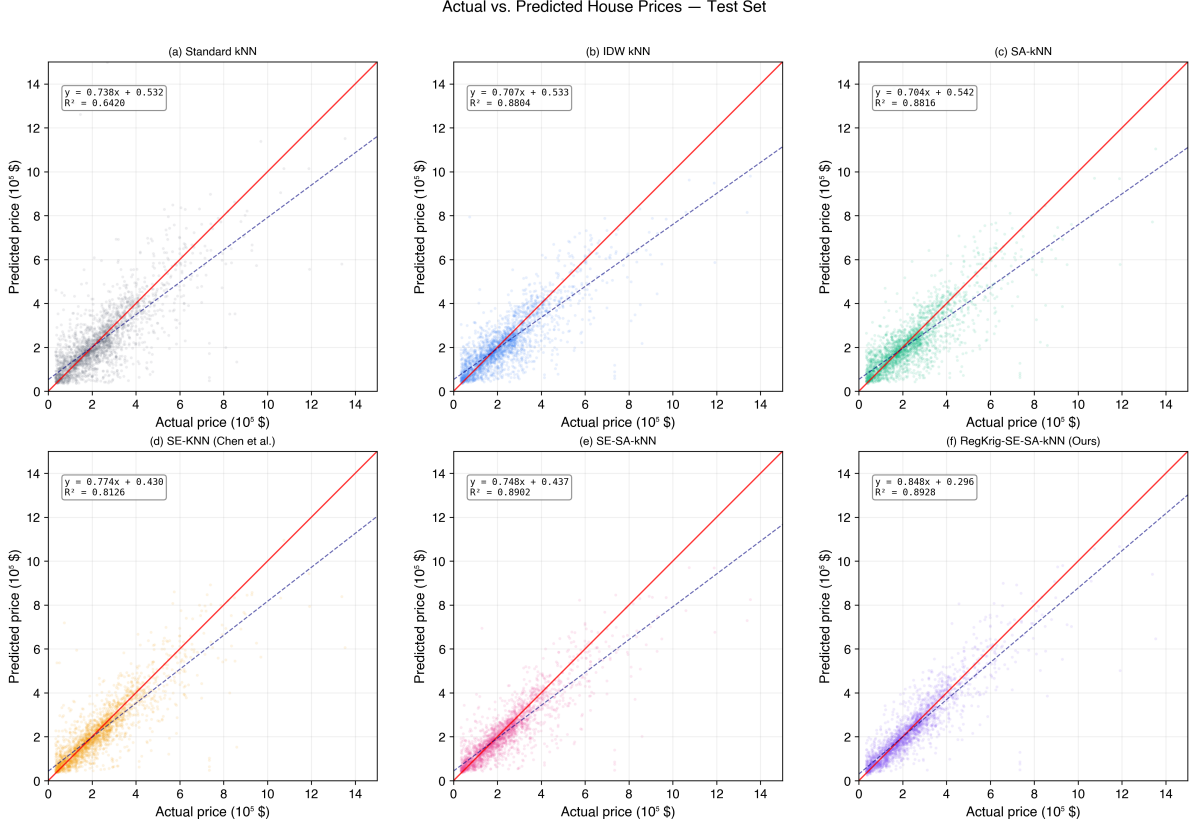


Figure 2: Actual vs. predicted house prices for all six models on the test set ($N_{\text{test}} = 2,476$). The red line is a 1:1 reference; the dashed blue line is a linear fit to the scatter. Some outlier predictions lie beyond the range.

Lee, and Chun (2026). Adding SA-kNN redundancy weights within the SE selection framework (SE-SA-kNN) reduces MAE further to \$52,606 (31.4% below the baseline). This validates the idea that the SE-selection step creates geometrically diverse neighbour sets in which inter-neighbour spatial correlations vary substantially, which then gives the redundancy correction power.

The regression-kriging correction provides the largest additional improvement: RegKrig-SE-SA-kNN reduces MAE by 8.4% relative to SE-SA-kNN, reaching \$48,201 (37.1% below the baseline). Notably, it also achieves the best R^2 (0.8928) and the lowest residual Moran’s I (0.1357), confirming that on this dataset, the kriging step both improves point accuracy and absorbs the remaining spatial structure in the residuals.

Figure 3 shows the RegKrig-SE-SA-kNN predicted prices (using 5-fold cross-validated predictions). The predicted spatial distribution closely mirrors the observed price map in Figure 1: the model correctly captures the north - south gradient, the concentration of low-value properties in inner city Columbus, and the high-value zones in the northern suburbs. Occasional overpredictions (the predicted maximum of \$12.25M compared to the observed maximum of \$2.18M) are likely due to a small number of properties with unusual attribute

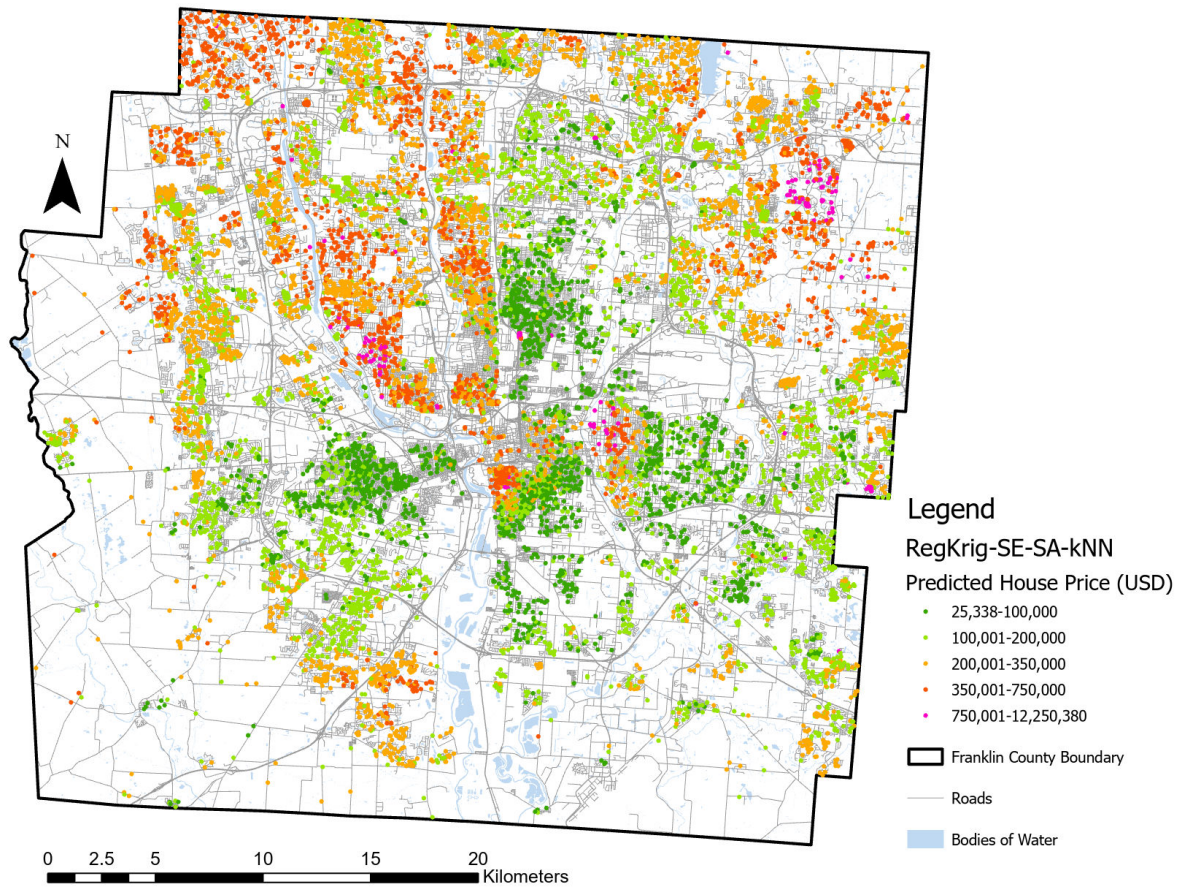


Figure 3: RegKrig-SE-SA-kNN predicted house prices across all 12,376 observations, displayed in the same price classes as Figure 1.

combinations that force the model to extrapolate at the feature-space boundary, which is consistent with the heavy right tail seen in RMSE (\$208,930) relative to MAE (\$48,201).

Figure 4 maps the absolute prediction error of RegKrig-SE-SA-kNN for each observation. The majority of observations show errors in the \$5,000–\$100,000 range, consistent with the overall MAE of \$48,201. A great number of residences are even predicted within \$20,000 of the correct value. Larger errors (\$100,000–\$500,000) are concentrated primarily in the urban centre Columbus and some suburbs, where property heterogeneity is greatest: in neighbourhoods with high price variance and mixed property types, the feature model is less able to distinguish structurally similar properties that sell at different prices. The spatial clustering of large errors, which is still visible (especially in three regions near the downtown core) even after the kriging correction, is consistent with the residual Moran's I of 0.1357.

From a critical GIS perspective, the geographic concentration of large errors in the urban core is not necessarily just a technical result, but could be more about the limits of attribute-based prediction. The 11 dwelling attributes describe the physical structure of each property, but they cannot encode the social geographies of neighbourhood reputation, disin-

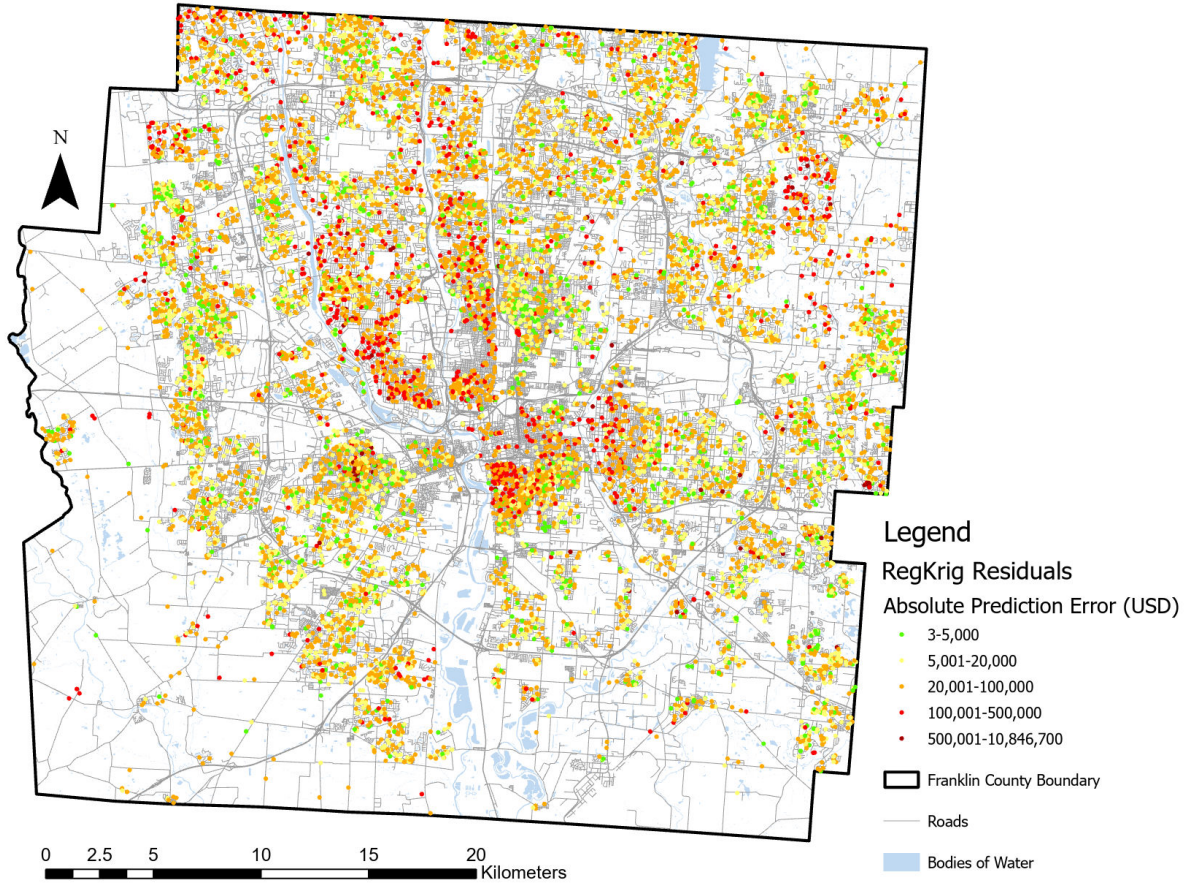


Figure 4: Absolute prediction error of RegKrig-SE-SA-kNN across all 12,376 observations. The majority of observations have smaller errors (green and orange); larger errors (red) are concentrated in the dense urban core and some northern high-value suburbs.

vestment, historical redlining, or gentrification that are well-documented drivers of urban price variance (Gamper-Rabindran and Timmins 2013). The model’s systematic difficulty in heterogeneous urban-core neighbourhoods reflects the idea that prices in these areas are partly determined by social processes that are not able to be captured by the feature set.

4.4 SA-kNN is approximately equivalent to IDW

We have also shown in this dataset (and believe this to be true in general, based on the following logic) that SA-kNN with the Gaussian proxy (8) is approximately equivalent to IDW kNN. The intuition is as follows. Under the Gaussian proxy,

$$C_{ij} = \exp(-\|\mathbf{s}_i - \mathbf{s}_j\|^2 / 2h^2),$$

so the redundancy score $r_i = \frac{1}{k-1} \sum_{j \neq i} C_{ij}$ is high precisely when neighbour i is geographically close to many other neighbours. In typical kNN neighbourhoods, geographic closeness is strongly correlated with feature-space closeness (due to spatial autocorrelation), which is

in turn correlated with small feature-space distance d_i . Therefore, $1 - r_i$ tends to be large for the same neighbours for which d_i is small. When combined with the d_i^{-1} term (from (6)), the result is a weighting scheme that is a smooth function of both feature-space and geographic distance, which is very similar to IDW.

Empirically, on this dataset, SA-kNN (\$63,897) and IDW (\$62,903) achieve nearly identical MAE (Table 1). This highlights why SA-kNN redundancy weights should be combined with SE-KNN’s blended selection to provide additional benefit. It would be worth exploring, however, if applying IDW on top of SE-KNN would have a similar effect.

4.5 Why regression-kriging reduces residual Moran’s I

The regression-kriging correction directly targets the residual spatial autocorrelation that the feature-based model cannot eliminate. Because the 11 dwelling attributes describe the individual property but not its neighbourhood context, spatial price gradients remain in the residuals. The kriging correction interpolates these gradients from training residuals and adds them back, producing the lowest residual Moran’s I of all models (0.1357; Table 1). Nonetheless, there is still plenty of residual spatial autocorrelation.

Figure 5 maps the property-level improvement from the kriging correction, that is, the reduction in absolute error from switching from SE-SA-kNN to RegKrig-SE-SA-kNN. Green points indicate properties where the kriging correction helped; orange and red points indicate properties where it hurt. Interestingly, the greatest clusters of improvements are also in the aforementioned areas near the urban core where the residuals of RegKrig-SE-SA-kNN are most clustered. This suggests that the areas where our prediction techniques struggle the most are also those where the kriging layer provides the most improvement. (A cartographic analysis, not shown here, confirms that SE-KNN and SE-SA-kNN indeed struggle the most in these areas.) The few extreme values at the tails of the improvement distribution (both very positive and very negative) are well-dispersed throughout the study area and reflect individual properties at the extremes of the price range where all models are quite inaccurate.

4.6 Comparison with Chen, Lee, and Chun (2026)

Chen, Lee, and Chun (2026) apply SE-KNN to a similar Franklin County dataset and report an MAE of approximately \$43,500 for SE-KNN. Our SE-KNN replication (MAE \$57,476) is higher, which, given that we have done our best to recreate their methodology, we assume must reflect differences in validity-code filtering, price floor, sample size, parameter search range, and random seed. Among other discrepancies, our tuning process selected a blended weight of $\lambda = 0.6$ rather than their $\lambda = 0.9$. Our proposed RegKrig-SE-SA-kNN algorithm nonetheless outperforms their reported SE-KNN compared to the relative baselines.

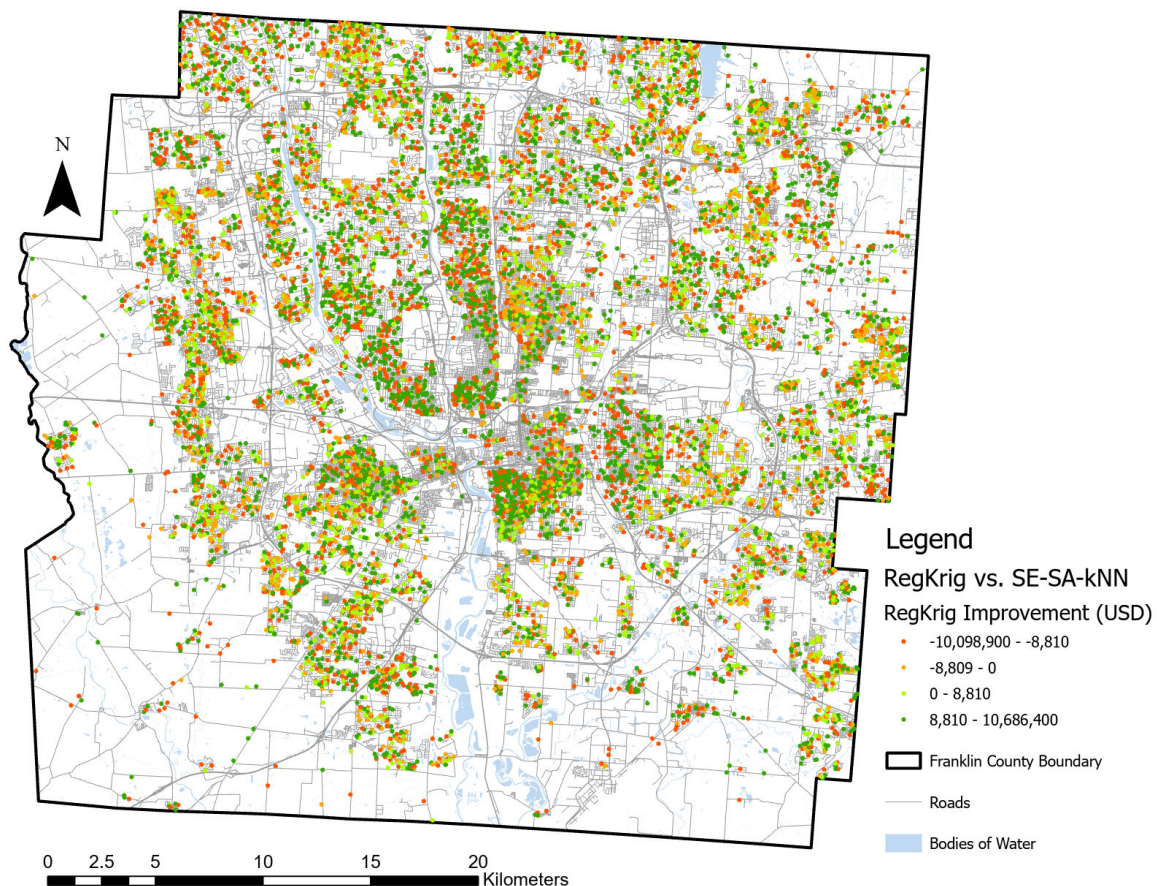


Figure 5: Property-level kriging improvement: the reduction in absolute prediction error when switching from SE-SA-kNN to RegKrig-SE-SA-kNN (positive = RegKrig closer to actual price; negative = SE-SA-kNN closer).

4.7 Overfitting.

One must ask if our development of RegKrig-SE-SA-kNN may be an example of overfitting. In particular, during the development of this project, we first proposed SA-kNN, before we incorporated SE-KNN from Chen, Lee, and Chun (2026), and then sought to improve on it. In this improvement process, we discovered the idea of regression kriging, and then implemented it on the Franklin County dataset, and found that it lead to great improvement. Although the theoretical basis of regression kriging is perfectly valid, we have certainly drifted away from a neutral methodology, instead making choices that optimize performance on the specific dataset which we have chosen to work with. Nonetheless, it is still likely that SA-based methods such as SE-SA-kNN and RegKrig-SE-SA-kNN will be effective and provide improved prediction results on other data sets; it is simply the case that this effect may not be as pronounced as the one seen in this paper. In general, one must always be careful and ensure that they are using methods that are appropriate for the dataset; SA-based methods are not always appropriate. It is important to first examine the structure of autocorrelation within a proposed dataset before applying the method. Similarly, SE-based methods are

only possible if there are relevant non-geographic features available to predict from. When this is not possible, simpler methods such as IDW and standard SA-kNN have the advantage.

4.8 Directions.

Future work could pursue several directions. First, it would be interesting to see if the convergence of IDW and C_{ij} -based SA-kNN holds for other datasets. Secondly, SA-kNN could be applied to classification tasks, where the redundancy weighting would affect class inclusion probability estimates (rather than a continuous prediction). Third, spatial cross-validation (Roberts et al. 2017) could provide a more conservative assessment of generalisation accuracy than the randomly-chosen training-validation-split used here (based on the methodology of Chen, Lee, and Chun (2026)); this is more theoretically well-backed and could help the methodology transfer better to other datasets.

References

- Acosta, Jorge, Ronny Vallejos, and Daniel A. Griffith (2018). “On the Effective Geographic Sample Size”. In: *Journal of Statistical Computation and Simulation* 88.10, pp. 1958–1975.
- Anselin, Luc (1995). “Local Indicators of Spatial Association — LISA”. In: *Geographical Analysis* 27.2, pp. 93–115.
- Atkinson, Peter M. (2004). “Spatially Weighted Supervised Classification for Remote Sensing”. In: *International Journal of Applied Earth Observation and Geoinformation* 5.4, pp. 277–291.
- Baldominos, Alejandro, Iván Blanco, Antonio José Moreno, Rubén Iturrarte, Óscar Bernárdez, and Carlos Afonso (2018). “Identifying Real Estate Opportunities Using Machine Learning”. In: *Applied Sciences* 8, p. 2321.
- Bivand, Roger S., Edzer Pebesma, and Virgilio Gómez-Rubio (2013). *Applied Spatial Data Analysis with R*. 2nd. Springer.
- Chen, Meifang, Changho Lee, and Yongwan Chun (2026). “A Machine Learning Approach Using Spatially Explicit K-Nearest Neighbors for House Price Predictions”. In: *ISPRS International Journal of Geo-Information* 15.1, p. 46. DOI: [10.3390/ijgi15010046](https://doi.org/10.3390/ijgi15010046).
- Clark, Philip J. and Francis C. Evans (1954). “Distance to Nearest Neighbor as a Measure of Spatial Relationships in Populations”. In: *Ecology* 35.4, pp. 445–453.
- Cliff, Andrew D. and J. Keith Ord (1973). *Spatial Autocorrelation*. Pion.
- (1981). *Spatial Processes: Models and Applications*. Pion.
- Cover, Thomas M. and Peter E. Hart (1967). “Nearest Neighbor Pattern Classification”. In: *IEEE Transactions on Information Theory* 13.1, pp. 21–27.
- Cressie, Noel A. C. (1993). *Statistics for Spatial Data*. Revised. John Wiley & Sons.
- Dudani, Sahibsingh A. (1976). “The Distance-Weighted k-Nearest-Neighbor Rule”. In: *IEEE Transactions on Systems, Man, and Cybernetics* 6, pp. 325–327.
- Fix, Evelyn and Joseph L. Hodges (1951). *Discriminatory Analysis—Nonparametric Discrimination: Consistency Properties*. Tech. rep. 4. Randolph Field, TX: USAF School of Aviation Medicine.
- Foresti, Loris, Alexei Pozdnoukhov, Devis Tuia, and Mikhail Kanevski (2010). “Extreme Precipitation Modelling Using Geostatistics and Machine Learning Algorithms”. In: *geoENV VII — Geostatistics for Environmental Applications*. Berlin: Springer, pp. 41–52.
- Fotheringham, A. Stewart, Chris Brunsdon, and Martin Charlton (2002). *Geographically Weighted Regression: The Analysis of Spatially Varying Relationships*. Wiley.
- Gamper-Rabindran, Shanti and Christopher Timmins (2013). “Does Cleanup of Hazardous Waste Sites Raise Housing Values? Evidence of Spatially Localized Benefits”. In: *Journal of Environmental Economics and Management* 65.3, pp. 345–360.

- Geary, Robert C. (1954). "The Contiguity Ratio and Statistical Mapping". In: *The Incorporated Statistician* 5.3, pp. 115–145.
- Getis, Arthur and J. Keith Ord (1992). "The Analysis of Spatial Association by Use of Distance Statistics". In: *Geographical Analysis* 24.3, pp. 189–206.
- Goodchild, Michael F. (2004). "The Validity and Usefulness of Laws in Geographic Information Science and Geography". In: *Annals of the Association of American Geographers* 94.2, pp. 300–303.
- Griffith, Daniel A. (2005). "Effective Geographic Sample Size in the Presence of Spatial Autocorrelation". In: *Annals of the Association of American Geographers* 95.4, pp. 740–760.
- Griffith, Daniel A., Kusum Agarwal, Meifang Chen, Changho Lee, Elizabeth A. Panetti, Kyungmin Rhyu, Lakshmi Venigalla, and Xiyue Yu (2022). "Geospatial Socio-Economic/Demographic Data: The Existence of Spatial Autocorrelation Mixtures in Georeferenced Data — Part I". In: *Transactions in GIS* 26, pp. 72–87.
- Harley, John Brian (1989). "Deconstructing the Map". In: *Cartographica* 26.2, pp. 1–20.
- Hastie, Trevor, Robert Tibshirani, Gareth James, and Daniela Witten (2021). *An Introduction to Statistical Learning*. 2nd. New York: Springer.
- Hengl, Tomislav, Gerard B. M. Heuvelink, and David G. Rossiter (2007). "About Regression-Kriging: From Equations to Case Studies". In: *Computers & Geosciences* 33.10, pp. 1301–1315.
- Hengl, Tomislav, Madlene Nussbaum, Marvin N. Wright, Gerard B. M. Heuvelink, and Benedikt Gräler (2018). "Random Forest as a Generic Framework for Predictive Modeling of Spatial and Spatio-Temporal Variables". In: *PeerJ* 6, e5518.
- Janowicz, Krzysztof, Song Gao, Grant McKenzie, Yingjie Hu, and Budhendra Bhaduri (2020). "GeoAI: Spatially Explicit Artificial Intelligence Techniques for Geographic Knowledge Discovery and Beyond". In: *International Journal of Geographical Information Science* 34.4, pp. 625–636.
- Jiao, Zhangnan and Ran Tao (2025). "Geographical Gaussian Process Regression: A Spatial Machine-Learning Model Based on Spatial Similarity". In: *Geographical Analysis* 57, pp. 507–520.
- Kopczewska, Katarzyna (2022). "Spatial Machine Learning: New Opportunities for Regional Science". In: *Annals of Regional Science* 68, pp. 713–755.
- Krige, Danie G. (1951). "A Statistical Approach to Some Basic Mine Valuation Problems on the Witwatersrand". In: *Journal of the Chemical, Metallurgical and Mining Society of South Africa* 52.6, pp. 119–139.
- Li, Wenwen (2020). "GeoAI: Where Machine Learning and Big Data Converge in GIScience". In: *Journal of Spatial Information Science* 20, pp. 71–77.
- Liu, Xiaojian, Ourania Kounadi, and Raúl Zurita-Milla (2022). "Incorporating Spatial Autocorrelation in Machine Learning Models Using Spatial Lag and Eigenvector Spatial Fil-

- tering Features". In: *ISPRS International Journal of Geo-Information* 11.4, p. 242. DOI: [10.3390/ijgi11040242](https://doi.org/10.3390/ijgi11040242).
- Matheron, Georges (1963). "Principles of Geostatistics". In: *Economic Geology* 58.8, pp. 1246–1266.
- Moran, Patrick A. P. (1950). "Notes on Continuous Stochastic Phenomena". In: *Biometrika* 37.1–2, pp. 17–23.
- Nikparvar, Behnam and Jean-Claude Thill (2021). "Machine Learning of Spatial Data". In: *ISPRS International Journal of Geo-Information* 10, p. 600.
- Odeh, Inakwu O. A., Alex B. McBratney, and David J. Chittleborough (1995). "Further Results on Prediction of Soil Properties from Terrain Attributes: Heterotopic Cokriging and Regression-Kriging". In: *Geoderma* 67.3–4, pp. 215–226.
- Openshaw, S. and P. J. Taylor (1979). "A Million or So Correlation Coefficients: Three Experiments on the Modifiable Areal Unit Problem". In: *Statistical Applications in the Spatial Sciences*. Ed. by N. Wrigley. Pion, pp. 127–144.
- Openshaw, Stan (1984). *The Modifiable Areal Unit Problem*. Vol. 38. Concepts and Techniques in Modern Geography. Norwich: Geo Books.
- Peterson, Leif E. (2009). "K-Nearest Neighbor". In: *Scholarpedia* 4.2, p. 1883.
- Razavi-Termeh, Seyed Vahid, Abolghasem Sadeghi-Niaraki, Seyedeh Razavi, and Soo-Mi Choi (2024). "Enhancing Flood-Prone Area Mapping: Fine-Tuning the K-Nearest Neighbors (KNN) Algorithm for Spatial Modelling". In: *International Journal of Digital Earth* 17, p. 2311325.
- Roberts, David R., Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Aroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann (2017). "Cross-Validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure". In: *Ecography* 40.8, pp. 913–929.
- Roscher, Ribana, Bastian Bohn, Marco F. Duarte, and Jochen Garcke (2020). "Explainable Machine Learning for Scientific Insights and Discoveries". In: *IEEE Access* 8, pp. 42200–42216.
- Shepard, Donald (1968). "A Two-Dimensional Interpolation Function for Irregularly-Spaced Data". In: *Proceedings of the 23rd ACM National Conference*, pp. 517–524.
- Snow, John (1855). *On the Mode of Communication of Cholera*. 2nd. London: John Churchill.
- Thiessen, Alfred H. (1911). "Precipitation Averages for Large Areas". In: *Monthly Weather Review* 39.7, pp. 1082–1089.
- Tobler, Waldo R. (1970). "A Computer Movie Simulating Urban Growth in the Detroit Region". In: *Economic Geography* 46, pp. 234–240.
- Vallejos, Ronny and Felipe Osorio (2014). "Effective Sample Size of Spatial Process Models". In: *Spatial Statistics* 9, pp. 66–92.